## Question 31 (a)

| Criteria | Marks |
|---|---|
| • Provides correct solution | 2 |
| • Equates the two expressions for the gradients, or equivalent merit | 1 |

***Sample answer:***

gradient $XP$ = gradient $PY$

$$\frac{0-2}{x-1}=\frac{y-2}{0-1}$$

$$(x-1)(y-2)=2$$

$$y-2=\frac{2}{x-1}$$

$$y=\frac{2}{x-1}+2$$

$$=\frac{2+2x-2}{x-1}$$

$$=\frac{2x}{x-1}$$

## Question 31 (b)

| Criteria | Marks |
|---|---|
| • Provides correct solution | 4 |
| • Solves $\dfrac{dA}{dx} = 0$ and determines the nature of stationary point at $x = 2$ | 3 |
| • Finds $\dfrac{dA}{dx}$, or equivalent merit | 2 |
| • Attempts to find the area in terms of $x$ | 1 |

***Sample answer:***

$$A = \frac{x^2}{x-1} \qquad \left[\text{from } A = \frac{1}{2}xy \text{ and } y = \frac{2x}{x-1}\right]$$

$$\begin{aligned}\frac{dA}{dx} &= \frac{(x-1)2x - x^2(1)}{(x-1)^2} \\ &= \frac{2x^2 - 2x - x^2}{(x-1)^2} \\ &= \frac{x^2 - 2x}{(x-1)^2} \\ &= \frac{x(x-2)}{(x-1)^2}\end{aligned}$$

$\dfrac{dA}{dx} = 0$ when $x = 0$ or $x = 2$

discard $x = 0$ since $x > 1$. (from question)

| $x$ | 1.5 | 2 | 3 |
|---|---|---|---|
| $\dfrac{dA}{dx}$ | $-3$ | 0 | $\dfrac{3}{4}$ |

Slope changes from negative to positive so

$x = 2$ is a minimum.

$$A = \frac{4}{2-1}$$

$\therefore$ Minimum area $= 4$ units$^2$